

Unit – 5 : Session – 5 : SLO – 2 | Polynomial and Rational Expression Operations**I. Questions & Answers**

1. Multiply $(x+2)(x+3)$ — $\text{expand}((x+2)*(x+3)) \rightarrow x^2 + 5x + 6$.
2. Factor $x^2 - 4$ — $\text{factor}(x^2 - 4) \rightarrow (x-2)*(x+2)$ (difference of squares).
3. Add $1/(x+1) + 1/(x-1)$ — together/simplify: combine over the common denominator $(x+1)(x-1) \rightarrow ((x-1) + (x+1)) / (x^2 - 1) = 2x/(x^2 - 1)$.

II. Code

1. `from sympy import symbols, expand, factor, simplify`
2. `x = symbols('x')`
3. `print(expand((x+2)*(x+3))) # x**2 + 5*x + 6`
4. `print(factor(x**2 - 4)) # (x - 2)*(x + 2)`
5. `print(simplify(1/(x+1) + 1/(x-1))) # 2*x/(x**2 - 1)`